

MATA37H3 — Week 15

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Final Exam Review: Comprehensive)

EXAMPLE 1 — Answer

False.

Common misconception: $a_n \rightarrow 0$ is *necessary* but not *sufficient* for convergence — the classic counterexample is the harmonic series $\sum \frac{1}{n}$, where $a_n \rightarrow 0$ but the series diverges.

EXAMPLE 2 — Answer

False.

Common misconception: A Ratio Test limit of 1 is **inconclusive** — it proves neither convergence nor divergence. A common exam trap treats $L = 1$ as automatic divergence.

EXAMPLE 3 — Answer

False.

Common misconception: It's the reverse: absolute convergence *implies* ordinary convergence, but “conditionally convergent” specifically means the series converges while $\sum |a_n|$ diverges — an absolutely convergent series is never called conditionally convergent.

EXAMPLE 4 — Answer

True.

Common misconception: Continuity on a closed interval guarantees Riemann integrability — this is a standard existence theorem, distinct from asking whether the integral has an elementary antiderivative.

EXAMPLE 5 — Answer

False.

Common misconception: The interval of convergence can be open, closed, or half-open depending on endpoint behavior — only the *radius* R is guaranteed; each endpoint $x = a \pm R$ must be checked separately by direct substitution.

EXAMPLE 6 — Answer

$$-\sqrt{4-x^2} + C$$

Common misconception: Using $u = 4 - x^2$, $du = -2x dx$, gives $\int x(4 - x^2)^{-1/2} dx = -\frac{1}{2} \int u^{-1/2} du = -\sqrt{u} + C$. A common error is mishandling the $-\frac{1}{2}$ factor from du .

EXAMPLE 7 — Answer

Converges to $\frac{1}{2}$

Common misconception: $\int_0^\infty e^{-2x} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{2}e^{-2x}\right]_0^b = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$. Forgetting the $\frac{1}{2}$ from the chain rule on e^{-2x} is the usual slip.

EXAMPLE 8 — Answer

≈ 5.146

Common misconception: $\Delta x = 1$; nodes 0, 1, 2, 3, 4 give f -values 0, 1, $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, 2; $T_4 = \frac{1}{2}[0 + 2(1) + 2(1.414) + 2(1.732) + 2] \approx 5.146$ — forgetting to double the interior nodes (only the endpoints stay single-weighted) is the most common Trapezoidal error.

EXAMPLE 9 — Answer

Converges

Common misconception: With $u = \ln x$, $\int_2^\infty \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^\infty u^{-2} du$, a convergent p -integral ($p = 2 > 1$). Forgetting to substitute the bounds ($x = 2 \Rightarrow u = \ln 2$, not $u = 2$) is a common slip.

EXAMPLE 10 — Answer

Diverges

Common misconception: The Divergence Test applies directly: $a_n = \frac{(-1)^n n}{n^2+1}$ does not tend to 0 in absolute value — $|a_n| = \frac{n}{n^2+1} \rightarrow 0$ actually, so re-examine: $|a_n| \rightarrow 0$, so the Divergence Test does NOT immediately apply; instead compare $|a_n| = \frac{n}{n^2+1}$ to $\frac{1}{n}$ via LCT ($\lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = 1$), and since $\sum \frac{1}{n}$ diverges, the series does not converge absolutely; the Alternating Series Test then shows the alternating series itself **converges conditionally**. The trap is stopping after checking $|a_n| \rightarrow 0$ and wrongly concluding absolute convergence without a comparison test.

EXAMPLE 11 — Answer

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{2^{n+1}}, R = 2$$

Common misconception: Write $\frac{1}{x} = \frac{1}{2+(x-2)} = \frac{1}{2} \cdot \frac{1}{1+\frac{x-2}{2}}$, then apply the geometric series with ratio $-\frac{x-2}{2}$. Forgetting the overall $\frac{1}{2}$ factor is the common error.

EXAMPLE 12 — Answer

$$1 + \frac{x}{3} - \frac{x^2}{9} + \frac{5x^3}{81} - \dots$$

Common misconception: Using $r = \frac{1}{3}$: $\binom{1/3}{1} = \frac{1}{3}$, $\binom{1/3}{2} = \frac{\frac{1}{3} \cdot (-\frac{2}{3})}{2} = -\frac{1}{9}$, $\binom{1/3}{3} = \frac{\frac{1}{3} \cdot (-\frac{2}{3}) \cdot (-\frac{5}{3})}{6} = \frac{5}{81}$. A common error is reusing the $\sqrt{1+x}$ ($r = \frac{1}{2}$) coefficients instead of

recomputing for $r = \frac{1}{3}$.

EXAMPLE 13 — Answer

$\lim_{n \rightarrow \infty} S_n = 2$, so $\sum a_n$ converges to 2

Common misconception: By definition, a series converges exactly when its sequence of partial sums converges, and the sum equals that limit — a frequent misconception is trying to reconstruct a_n first rather than recognizing that the limit of S_n IS the answer directly.